



Uveal melanoma incidence and survival analysis in Portugal between 2013 and 2022

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ABSTRACT

Purpose: This study aims to provide a comprehensive overview of the clinical characteristics and epidemiology of uveal melanoma (UM) in the Portuguese population, evaluated at the National Reference Centre (NRC).

Methods: A prospective observational study was conducted, involving patients consecutively diagnosed with UM at the Portuguese NRC between July 2013 and December 2022. The study collected data on demographic and tumour characteristics, clinical staging according to the American Joint Committee on Cancer (AJCC), treatment approaches, local disease control, patient survival, and the occurrence of distant metastases.

Results: The study included a total of 316 patients, 53.8% female. The mean age at diagnosis was 61.8 ± 14.2 years, and 75.0% of patients presented with symptoms. The mean annual age-adjusted incidence of uveal melanoma in Portugal between 2014 and 2022 was 2.4 cases per million (95% confidence interval [CI]: 2.1–2.8). For choroidal/ciliary body tumours, the overall cumulative survival and distant metastases-free survival (DMFS) rates at 5 years were 84.9% (95% CI: 78.7–91.1) and 79.4% (95%CI: 72.8–86.0), respectively. Notably, higher AJCC stages at presentation, the need for enucleation, and increased tumour thickness were associated with lower DSS and DMFS rates.

Conclusion: This study represents the most extensive analysis of UM epidemiology within the Portuguese population. The findings underscore the importance of early diagnosis and treatment in UM, as lower AJCC stages and smaller tumour thickness at diagnosis correlate with improved DSS and DMFS.

1. Introduction

Although recognized as a rare condition, ocular melanoma is the most common ocular primary malignancy, representing 5% of all melanomas [1,2]. Among ocular melanomas, nearly 85% occur in the uveal tract [1,2], of which 85–90% arise from the choroid, followed by the ciliary body (5–8%) and the iris (3–5%) [2,3–6].

The incidence of UM in Europe has been reported as ranging from 2 cases per million per year in Southern Europe to over 8 cases per million per year in Northern countries [7]. In Asia and Africa the reports point to a significantly lower incidence rate, of less than 1 case per million per

year [6,8].

Advanced age is related to higher incidence rates, with both mean and median ages at diagnosis of approximately 62 years old [6,9–11]. Regarding incidence by gender, there is not a true consensus, with variable results according to different studies [6].

Along with location, tumour size, extrascleral extension and associated complications will determine clinical presentation, with most patients complaining of decreased visual acuity and blurred vision; about 30% are asymptomatic at presentation [6,12–15].

Clinical characteristics as older age at presentation, male gender, larger tumour basal diameter, increased tumour thickness, ciliary body

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